

PyTorch Dataset & DataLoader Learning Workflow

1. Core Concepts to Extract from Documentation

- Dataset must implement two methods: `__len__()` and `__getitem__(idx)`.
- `__len__` returns the total number of samples.
- `__getitem__` returns a single training sample given an index.
- DataLoader wraps a Dataset and handles batching, shuffling, and parallel loading.

2. Phase 1 — Build a Toy Dataset (No Files)

- Create a dataset that returns (x, x^2) .
- Store numbers internally in the dataset.
- Implement `__len__` and `__getitem__`.
- Test by calling `dataset[0]`, `dataset[1]`, etc.

3. Phase 2 — Connect Dataset to DataLoader

- Wrap the dataset using DataLoader.
- Experiment with `batch_size` and `shuffle`.
- Iterate through batches and inspect shapes and structure.

4. Phase 3 — Structured Dataset (Tabular Data)

- Load tabular data using pandas.
- Split features and labels.
- Convert them to tensors.
- Return (x_tensor, y_tensor) from `__getitem__`.

5. Phase 4 — File-Based Dataset

- Store file paths and labels.
- Load files lazily inside `__getitem__`.
- Preprocess data and convert to tensors.
- Return processed samples.

6. Phase 5 — Add Transforms

- Add a transform argument to the Dataset class.
- Apply preprocessing or augmentation inside `__getitem__`.
- This creates modular and reusable pipelines.

7. Phase 6 — Advanced DataLoader Features

- Experiment with `num_workers` for parallel loading.
- Use `collate_fn` for custom batching.
- These are important for NLP, detection, and variable-length inputs.

8. Phase 7 — Convert a Keras Generator to PyTorch

- Keras Sequence maps closely to PyTorch Dataset.
- Extract sample logic from the generator.
- Move batching logic to PyTorch DataLoader.

9. Practice Project

- Create a `CSVImageDataset`.
- Load image paths from a CSV file.
- Load and preprocess images.
- Return tensors suitable for training.

10. Debugging Strategy

- Always test dataset indexing first: `dataset[0]`, `dataset[1]`.
- Then inspect batches using `next(iter(loader))`.
- Print shapes, types, and ranges of tensors.

11. Mental Model

- Dataset produces one sample.
- DataLoader builds batches from samples.
- The model consumes batches.

12. Suggested Learning Timeline

- Day 1: Dataset basics.
- Day 2: DataLoader and batching.
- Day 3: File-based datasets.
- Day 4: Transforms and multiprocessing.
- Day 5: Convert a Keras dataloader.